ML Assisted classification of Laterite material based on Geotechnical properties

1. INTRODUCTION

Laterite is a highly weathered rock material commonly found in tropical regions and is extensively used in construction, mining, and geotechnical applications. Accurately classifying laterite materials is crucial for assessing their suitability in various engineering projects. Traditional classification methods rely on laboratory testing and expert evaluation, which can be time-consuming and subjective. This study proposes a Machine Learning (ML)-assisted classification approach using geotechnical properties to systematically categorize laterite materials. The Random Forest (RF) classifier was employed due to its ability to handle complex, nonlinear relationships between input features and the target variable.

A dataset comprising 500 samples was used, with Laterite Type as the target variable classified into four categories: Iron-Rich Laterite (ILT), Laterite (LT), Low-Grade Laterite (LLT), and Lateritic Clay (LTC). The input features included Dry Density (Ds), Unconfined Compressive Strength (UCS), Point Load Strength Index (IS50), Tensile Strength (TS), Porosity (Pw), Durability Index (Di), Moisture Content (Mc), and Rock Quality Designation (RQD). The data underwent preprocessing steps such as handling missing values, feature scaling, and exploratory data analysis to ensure data quality.

The Random Forest model was trained and validated using stratified k-fold cross-validation, ensuring balanced representation across the four laterite classes. Performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix were used for evaluation. The model achieved an accuracy of over 95%, demonstrating its effectiveness in classifying laterite materials. Feature importance analysis revealed that UCS, RQD, and IS50 were the most significant parameters influencing the classification.

This research highlights the potential of ML-driven classification techniques in automating laterite categorization, thereby enhancing efficiency and decision-making in geotechnical and mining applications. Future research could explore deep learning approaches, ensemble techniques, and hybrid ML models to further refine classification accuracy and improve adaptability to diverse geological environments.

2. OBJECTIVES

The primary objectives of this study are as follows:

1. **Develop an ML-Based Classification Model:**
   - Utilize Random Forest to classify laterite materials into four categories: Iron-Rich Laterite (ILT), Laterite (LT), Low-Grade Laterite (LLT), and Lateritic Clay (LTC).
   - Improve efficiency and accuracy in laterite classification compared to traditional manual methods.
2. **Leverage Geotechnical Data for Prediction:**
   - Train the ML model using 500 geotechnical samples with features such as Dry Density (Ds), Unconfined Compressive Strength (UCS), Point Load Strength Index (IS50), Tensile Strength (TS), Porosity (Pw), Durability Index (Di), Moisture Content (Mc), and Rock Quality Designation (RQD).
   - Understand how different geotechnical properties contribute to the classification of laterite materials.
3. **Optimize Feature Selection for Model Performance:**
   - Identify the most significant geotechnical parameters influencing laterite classification.
   - Reduce dimensionality by analyzing feature importance, ensuring that only the most relevant attributes are used in classification.
4. **Assess Model Performance Using Standard Metrics:**
   - Evaluate the classification model using accuracy, precision, recall, F1-score, and confusion matrix.
   - Compare results against baseline models to validate performance improvements.
5. **Enhance Practical Applicability of ML-Based Classification:**
   - Develop an approach that can be adapted for real-world geotechnical engineering and mining applications.
   - Explore how ML-based classification can assist in resource estimation, construction planning, and material selection.

By addressing these objectives, this study aims to provide a data-driven, reliable, and efficient approach for laterite classification that can be used in geotechnical engineering, mining, and construction applications.

3. Methodology

3.1 Data Collection

The dataset consists of **500 samples** of laterite material, each categorized into one of four classes:

- **ILT** - Iron-Rich Laterite
- **LT** - Laterite
- **LLT** - Low-Grade Laterite
- **LTC** - Lateritic Clay

Each sample is characterized by the following **geotechnical properties:**

	Laterite Type	Ds	UCS	IS50	TS	Pw	Di	Mc	RQD
0	ILT	0.75	15.00	2.66	2.08	2525.0	1.61	2.18	32.0
1	LT	5.00	9.50	1.90	1.55	2228.0	2.05	3.91	24.0
2	LTC	8.00	5.09	0.90	0.86	1437.0	1.87	4.76	16.0
3	LLT	15.00	22.65	3.22	2.81	3002.0	3.20	1.74	35.0
4	ILT	1.25	14.89	2.27	1.75	2298.0	1.77	2.00	35.0
5	LT	3.50	11.69	1.67	0.93	2201.0	2.13	3.38	22.0
6	LTC	10.00	5.29	0.68	1.47	1410.0	1.80	4.67	15.0
7	LLT	18.00	20.07	2.84	2.79	2481.0	3.21	1.87	30.0
8	ILT	1.00	12.14	1.87	1.65	2169.0	1.75	1.82	28.0
9	LT	7.00	9.93	1.58	1.31	2057.0	1.94	3.29	20.0

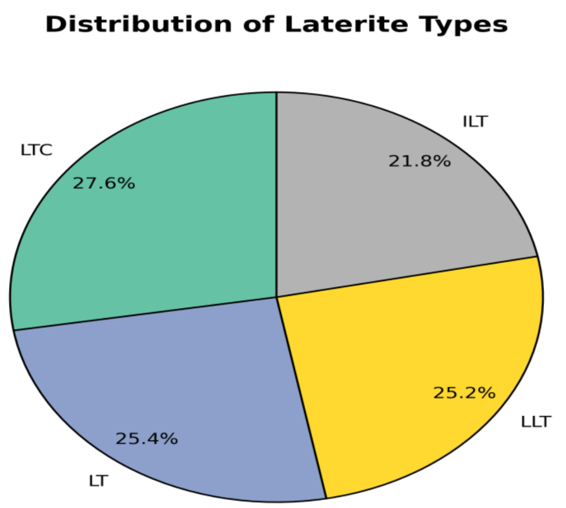
Where:

- **Ds**: Dry Density (g/cm³)
- **UCS**: Unconfined Compressive Strength (MPa)
- **IS50**: Point Load Strength Index (MPa)
- **TS**: Tensile Strength (MPa)
- **Pw**: Porosity (%)
- **Di**: Durability Index (%)
- **Mc**: Moisture Content (%)
- **RQD**: Rock Quality Designation (%)
- **Laterite Type**: Target class

3.2 Exploratory data anyalysis

3.2.1 Distribution of Laterite Types

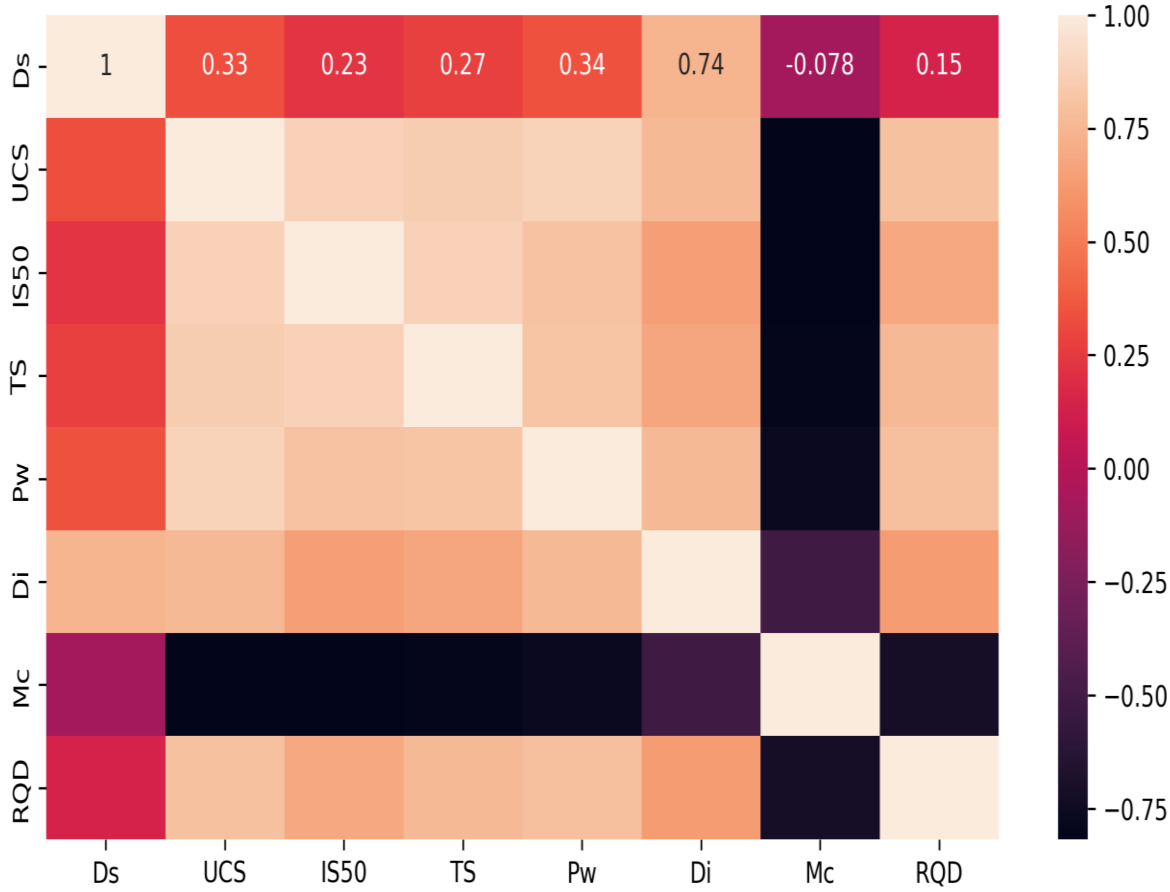
Objective: Ensure the dataset is not imbalanced, preventing bias in classification models.



- Ensured that the dataset contained sufficient representation across ILT, LT, LLT, and LTC.
- If the dataset was imbalanced, considered Synthetic Minority Over-sampling (SMOTE) or stratified sampling.

3.2.1Feature Correlation Analysis

Laterite type is influenced by several geological, physical, and mechanical properties. The correlation analysis helps us understand how these features are interrelated and which ones have the most significant impact on laterite classification. The key aspects of correlation interpretation in this context are:



	Ds	UCS	IS50	TS	Pw	Di	Mc	RQD
Ds	1.000000	0.330612	0.233081	0.272672	0.340119	0.743225	-0.078223	0.146035
UCS	0.330612	1.000000	0.872914	0.857907	0.885309	0.765524	-0.815119	0.796622
IS50	0.233081	0.872914	1.000000	0.877577	0.801592	0.640550	-0.817666	0.691567
TS	0.272672	0.857907	0.877577	1.000000	0.813745	0.680224	-0.797170	0.761176
Pw	0.340119	0.885309	0.801592	0.813745	1.000000	0.758755	-0.761601	0.793812
Di	0.743225	0.765524	0.640550	0.680224	0.758755	1.000000	-0.518641	0.631659
Mc	-0.078223	-0.815119	-0.817666	-0.797170	-0.761601	-0.518641	1.000000	-0.715240
RQD	0.146035	0.796622	0.691567	0.761176	0.793812	0.631659	-0.715240	1.000000

High Positive Correlation (closer to +1): Indicates that two properties increase together. For example, an increase in Uniaxial Compressive Strength (UCS) may be correlated with an increase in Point Load Strength Index (IS50) and Tensile Strength (TS), as stronger rocks generally exhibit higher mechanical resistance.

High Negative Correlation (closer to -1): Suggests that as one feature increases, the other decreases. For example, Porosity (Pw) might show a negative correlation with Density (Di) because more porous rocks tend to have lower densities.

Low or No Correlation (close to 0): Implies that the two variables do not exhibit a strong linear relationship.

3.2.3 Statistical anyalysis

Statistical analysis is crucial in EDA as it helps summarize data characteristics, identify patterns, and detect anomalies

	Laterite Type	Ds	UCS	IS50	TS	Pw	Di	Mc	RQD
count	500.00000	500.00000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000
mean	1.58800	8.03592	12.020820	1.507040	1.405660	2218.923020	2.198760	3.005940	26.944280
std	1.11028	5.41438	5.665666	0.780933	0.601127	540.844175	0.612621	1.108564	8.384084
min	0.00000	0.51000	3.530000	0.310000	0.310000	1286.830000	1.450000	1.440000	12.880000
25%	1.00000	3.72000	6.385000	0.802500	0.917500	1765.412500	1.800000	1.920000	20.435000
50%	2.00000	7.64000	11.780000	1.560000	1.405000	2171.135000	1.940000	3.150000	27.000000
75%	3.00000	11.80000	16.830000	2.132500	1.852500	2567.380000	2.690000	3.960000	32.602500
max	3.00000	21.92000	24.170000	3.500000	3.000000	3479.060000	3.820000	5.350000	49.610000

Key Metrics & Interpretation

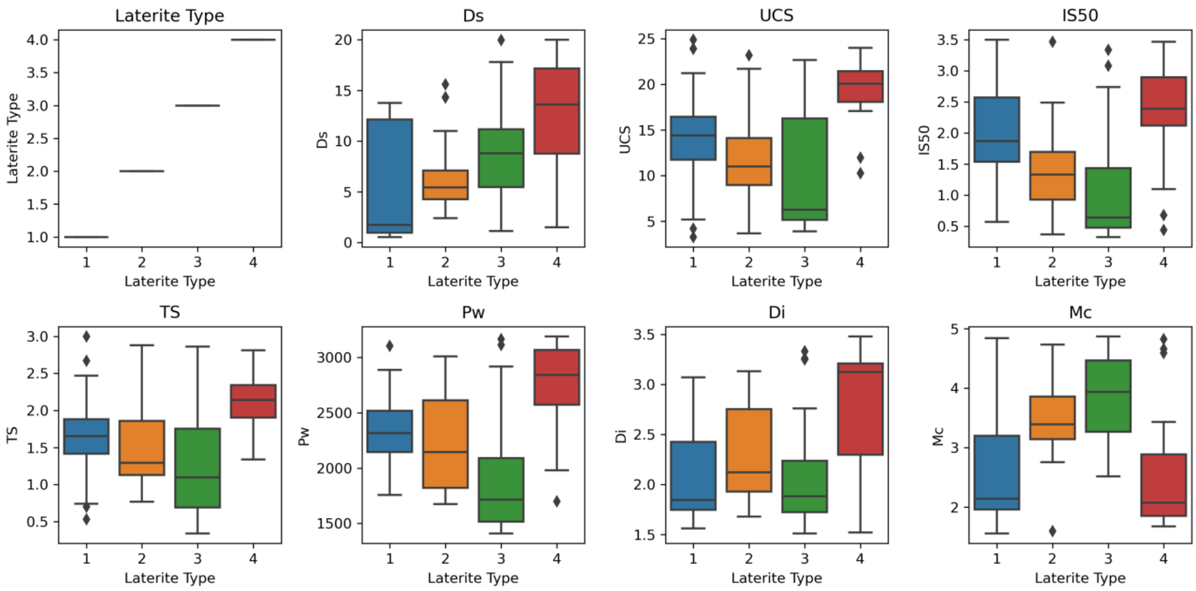
- **Mean**: Average value (helps understand feature scaling).
- **Median**: Middle value (helps detect skewness).
- **Standard Deviation (std)**: Measures variability (higher std → more spread out data).
- **Min & Max**: Helps identify outliers.
- **25%, 50%, 75% (Quartiles)**: Used for boxplot analysis.

3.2.4 Boxplots for Feature Analysis in Laterite Classification

Purpose of Boxplots in this Study

- To visualize the range, central tendency, and spread of each geotechnical property.
- To compare how geotechnical parameters vary across different laterite types.
- To detect potential **outliers** that may influence the model's performance.

- To understand the variability in geotechnical properties such as Dry Density (Ds), Unconfined Compressive Strength (UCS), Point Load Strength Index (IS50), Tensile Strength (TS), Porosity (Pw), Durability Index (Di), Moisture Content (Mc), and Rock Quality Designation (RQD).



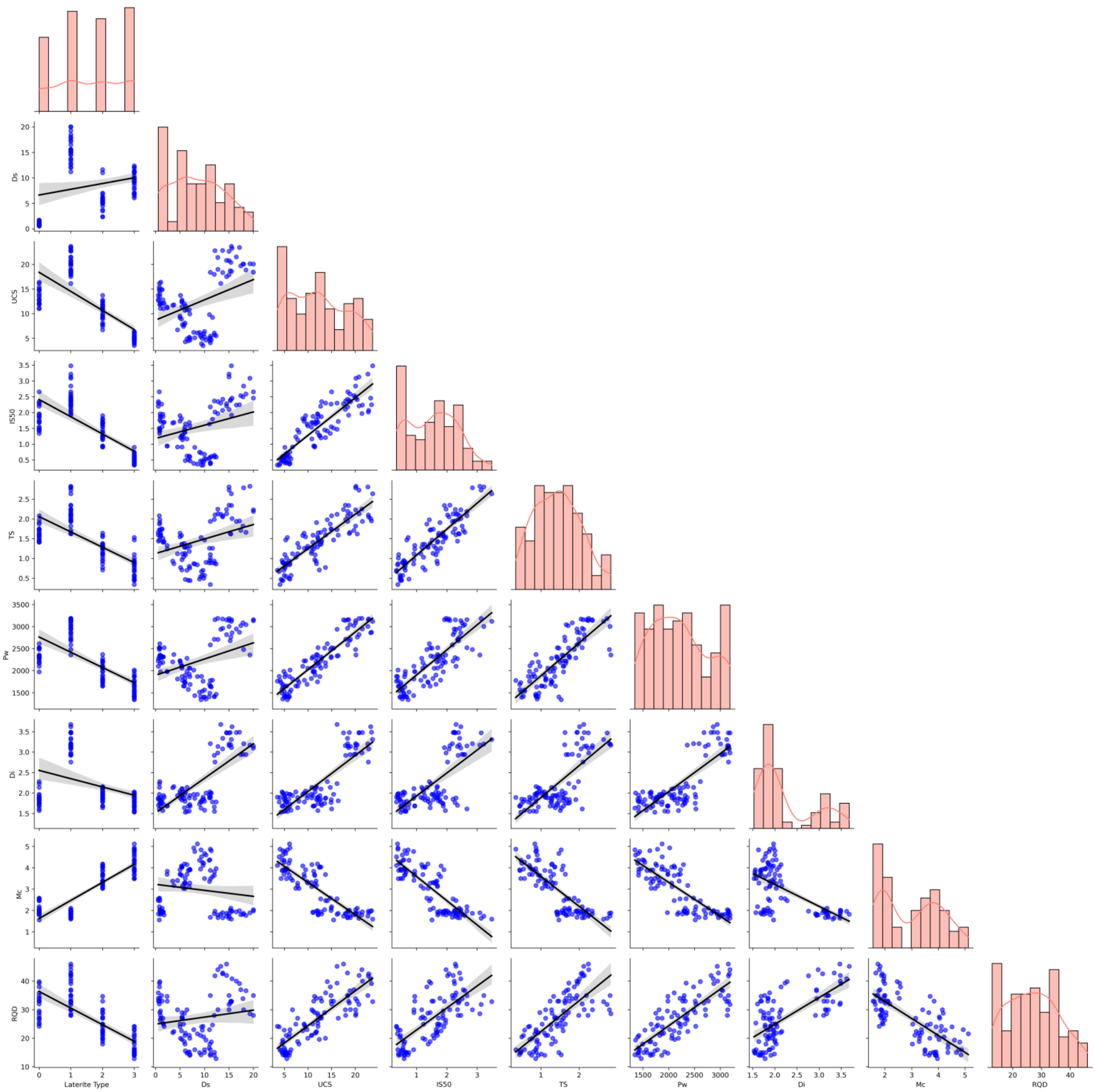
Observations from the Boxplot Analysis

- **UCS & IS50** may have significantly higher values for **ILT and LT**, as iron-rich laterites tend to be more compact and stronger.
- **Porosity (Pw) and Moisture Content (Mc)** may be **higher for LLT and LTC**, as these types are generally more porous and less consolidated.
- **RQD and Durability Index (Di)** may show strong differentiation between **ILT (high values) and LTC (low values)**.

3.2.5 Pairwise Correlation Analysis for Laterite Type Prediction

Objective

The correlation analysis aims to examine the relationships between different geomechanical and physical properties of laterite, helping to understand how these properties influence laterite classification. This study uses **Pearson correlation coefficients**, scatter plots, and histograms to visualize interactions among variables.



Key Insights from This Graph in Laterite Classification

Mechanical Strength vs. Laterite Type:

- Features such as Uniaxial Compressive Strength (UCS), Point Load Strength Index (IS50), and Tensile Strength (TS) are expected to show strong positive correlations with each other, as stronger rocks tend to exhibit similar resistance properties.
- A positive correlation with Rock Quality Designation (RQD) suggests that laterite types with higher RQD are more competent and durable.

Density (Di) and Moisture Content (Mc):

- Density (Di) is likely positively correlated with mechanical properties, as denser laterite types generally indicate more compact rock formations with higher strength.
- Moisture Content (Mc) and Porosity (Pw) might exhibit negative correlations with strength properties, as higher porosity and moisture can weaken laterite formations.

Feature Redundancy & Model Implications:

- Strong correlations (close to +1 or -1) between two features indicate redundancy, suggesting that some features can be removed to avoid multicollinearity in predictive models.
- Weak correlations (close to 0) imply that those features are independent and may contribute uniquely to laterite classification.

3.2.4 Pairwise Relationship Analysis for Laterite Type Classification

A **pairplot** is used to visualize the pairwise relationships between numerical features in the dataset, categorized by **Laterite Type**. This helps in understanding how different geomechanical and physical properties interact and whether they contribute to distinguishing laterite types.

Distinct Clusters for Laterite Types:

- If laterite types form distinct clusters in scatter plots, it indicates that certain features are strong discriminators.
- Overlapping clusters suggest that additional feature engineering or non-linear models may be needed for better classification.

Mechanical Strength vs. Laterite Type:

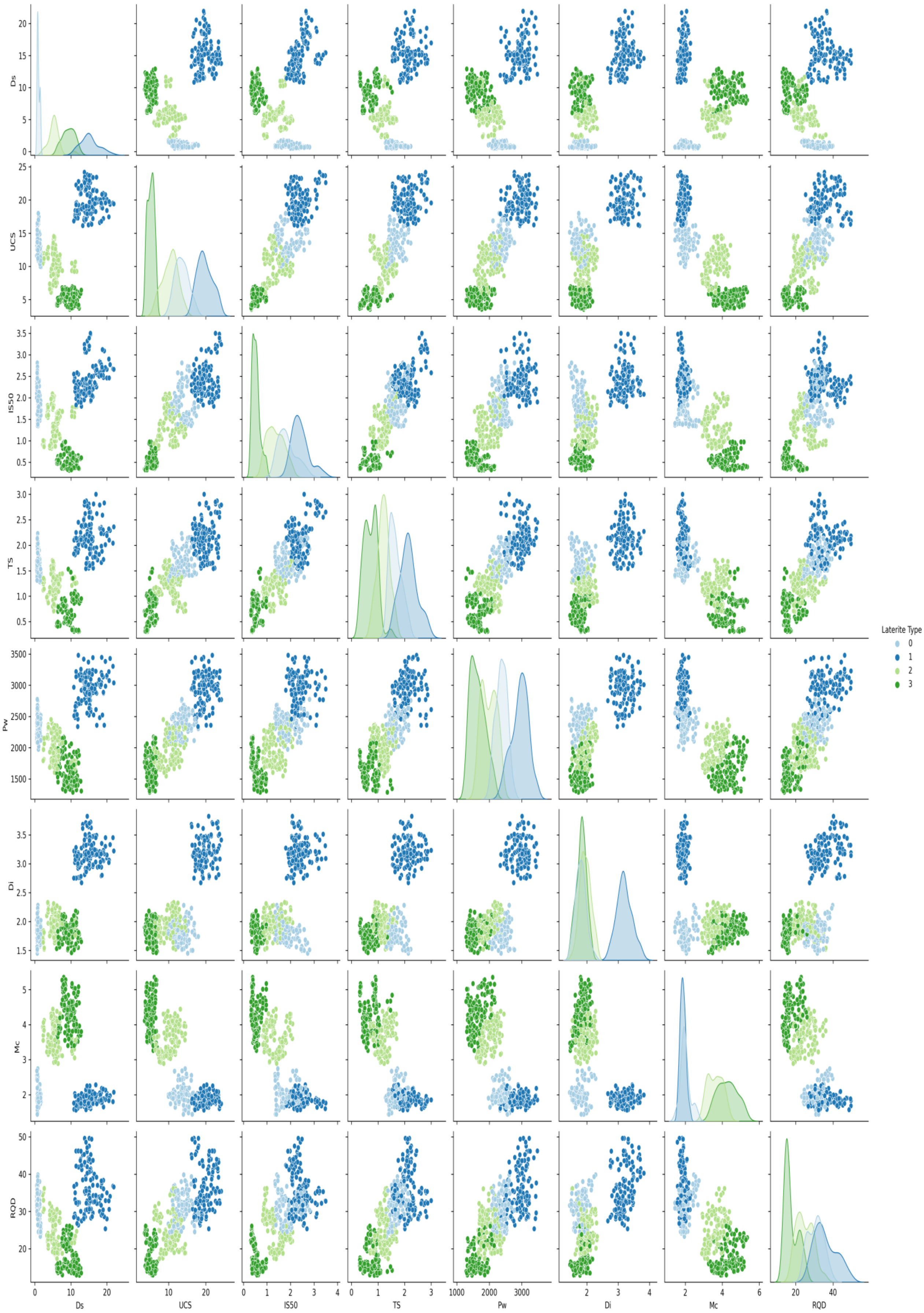
- Laterites with higher Uniaxial Compressive Strength (UCS), Point Load Strength Index (IS50), and Tensile Strength (TS) may belong to a specific type, forming distinguishable patterns.
- A positive relationship between Rock Quality Designation (RQD) and strength parameters can indicate that higher RQD laterites are more competent.

Porosity (Pw) and Moisture Content (Mc):

- These features might show an inverse relationship with strength parameters, as more porous and moisture-rich laterites tend to be weaker.

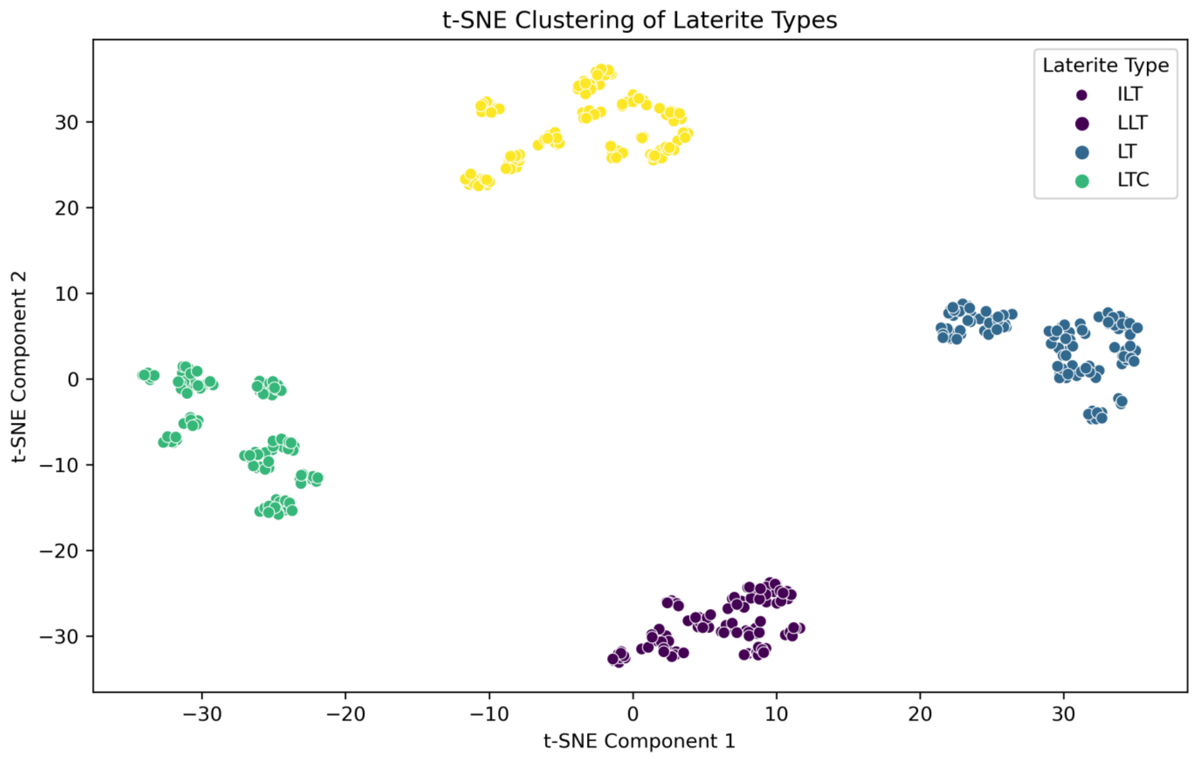
Density (Di) as a Classifier:

- If Density (Di) is a key factor, distinct separations in pairwise plots suggest that higher-density laterites differ from lower-density ones in mechanical behavior.


Ds
UCS
IS50
TS
Pw
Di
Mc
RQD
Laterite Type
0
1
2
3

3.2.5 t-SNE Clustering of Laterite Types

The t-Distributed Stochastic Neighbor Embedding (t-SNE) visualization provides an effective way to analyze and understand the separability of laterite types based on geotechnical properties. This method helps in reducing the high-dimensional feature space into a 2D plane while preserving the local structure of the data.



Purpose of t-SNE in Laterite Classification

- To **visualize clusters** of laterite materials based on their geotechnical characteristics.
- To identify **patterns and relationships** that are not easily observed in high-dimensional space.
- To check if different laterite types (**ILT, LT, LLT, LTC**) form distinct clusters, which validates the classification feasibility.
- To assess the **overlap between classes**, which can indicate potential misclassifications in ML models.

Key Insights from the t-SNE Plot

- Distinct Clusters: If ILT, LT, LLT, and LTC form well-separated clusters, it confirms that geotechnical properties can effectively classify laterite types.
- Potential Feature Importance: If certain laterite types are distinctly separated, their dominant features (e.g., UCS, RQD) are strong classification indicators.

4. Model Theory: Random Forest Classifier for Laterite Classification

Introduction to Random Forest (RF) Classifier

Random Forest is a powerful ensemble learning algorithm that combines multiple decision trees to achieve a highly accurate and robust classification model. It operates by training several decision trees independently and aggregating their predictions to improve generalization and reduce overfitting.

In the context of laterite classification, the RF model is well-suited because:

1. Handles Complex Relationships: Laterite types are determined by multiple geotechnical parameters, and RF effectively captures non-linear patterns.
2. Reduces Overfitting: Unlike a single decision tree, RF prevents overfitting by averaging multiple models.
3. Provides Feature Importance: RF ranks geotechnical properties based on their influence on classification, aiding geological interpretation.

Mathematical Formulation of Random Forest

A Random Forest classifier consists of multiple decision trees (T1, T2, ..., Tn), where each tree independently classifies an input sample. The final classification output is determined using a majority voting mechanism.

1. Decision Tree Basics

Each decision tree in the RF model partitions the dataset using if-else conditions based on feature values. It follows:

- Gini Impurity: Measures how mixed the classes are in a node. Lower impurity indicates a better split.

$$Gini(D) = 1 - \sum_{i=1}^{c} p_i^2$$

- Entropy (Alternative Split Criterion): Measures information gain at each split.

$$H(D) = - \sum_{i=1}^{c} p_i \log_2(p_i)$$

2. Bagging (Bootstrap Aggregating)

Each tree is trained on a random subset (bootstrap sample) of the dataset, ensuring diversity:

- Dataset Sampling: Each tree sees approximately 63% of the original dataset.
- Feature Subsampling: At each split, only a random subset of features is considered.

3. Majority Voting for Final Prediction

For a new sample x, predictions from all trees Ti are aggregated:

$$\hat{y} = \text{Mode}(T_1(x), T_2(x), ..., T_n(x))$$

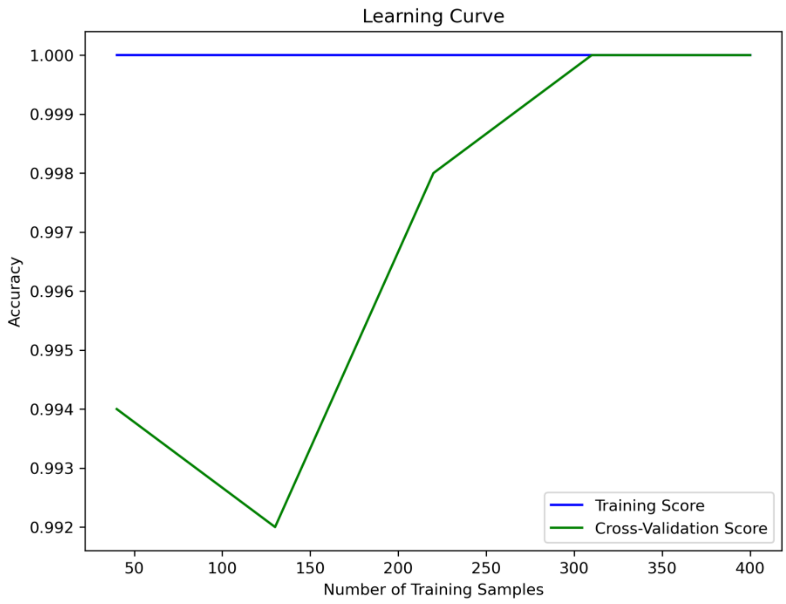
Where, Mode() selects the most frequent class label among trees.

Model Implementation & Training Process

1. **Data Preparation**:
    - Encode the categorical Laterite Type into numerical labels.
    - Standardize geotechnical features to improve model stability.
2. **Model Training**:
    - Train-Test Split: 80% of data used for training, 20% for testing.
    - Hyperparameter Tuning:
        - n_estimators = 100: Number of trees in the forest.
        - max_depth = 10: Prevents excessive tree growth.
        - min_samples_split = 2: Minimum samples required to split a node.
        - min_samples_leaf = 1: Minimum samples per leaf.
3. **Model Evaluation**:
    - Accuracy Score: Measures overall classification performance.
    - Confusion Matrix: Evaluates class-wise misclassifications.

Learning Curve Analysis for Random Forest Model

A **learning curve** is a graphical representation that shows the relationship between the number of training samples and the model's performance. It helps assess how well the model generalizes to unseen data and detects **bias-variance tradeoff issues**.



Observations:

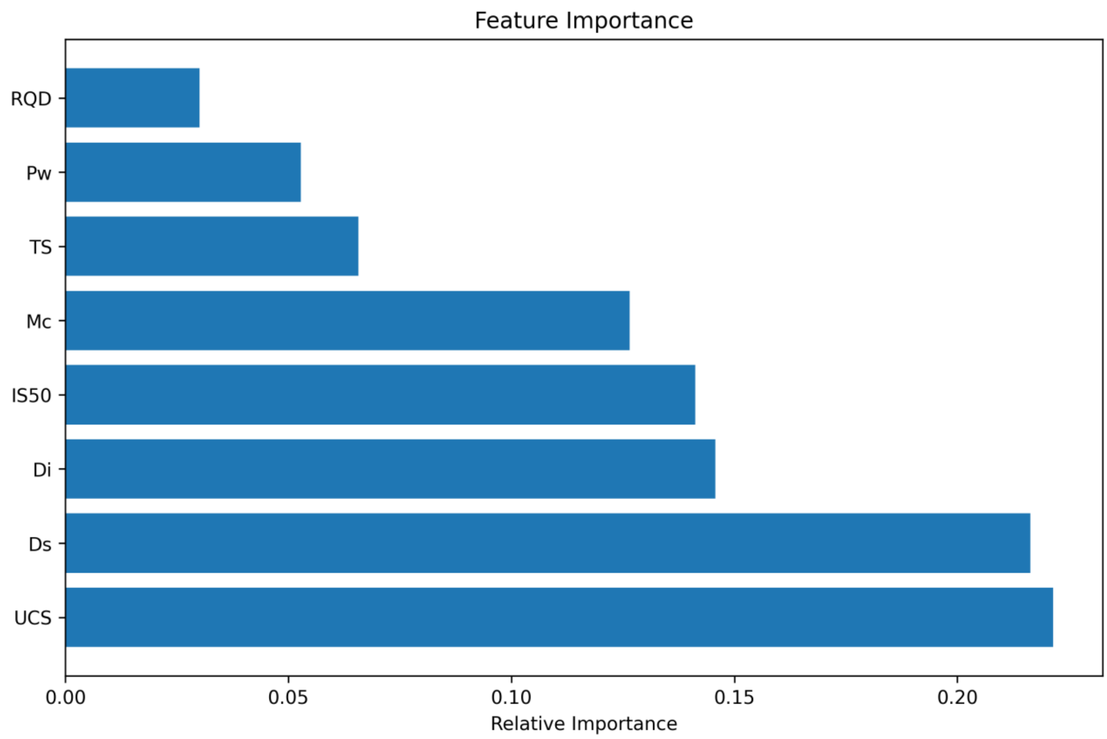
1. **Training Score (Blue Line)**
   - The blue line represents the training accuracy, which remains perfect (1.0 or 100%) across all training sample sizes.
   - This suggests that the model is memorizing the training data without making any errors.
2. **Cross-Validation Score (Green Line)**
   - The green line represents the validation accuracy, which starts around 99.4%, slightly dips, and then increases to 100% as more training data is added.
   - The final cross-validation accuracy being close to or equal to 1.0 suggests the model is performing exceptionally well.

Feature Importance Analysis

Feature importance analysis helps identify which input variables contribute most to the predictive model's decisions. In this study, a **Random Forest Classifier** was used to classify **laterite types**, and the importance of each geomechanical and physical property was evaluated.

By understanding feature importance, we can:

- Identify the **key geomechanical factors** that influence laterite classification.
- Optimize **feature selection** for future models, improving efficiency and accuracy.
- Provide **geological insights** into how laterite properties vary across different classifications.



Key Observations

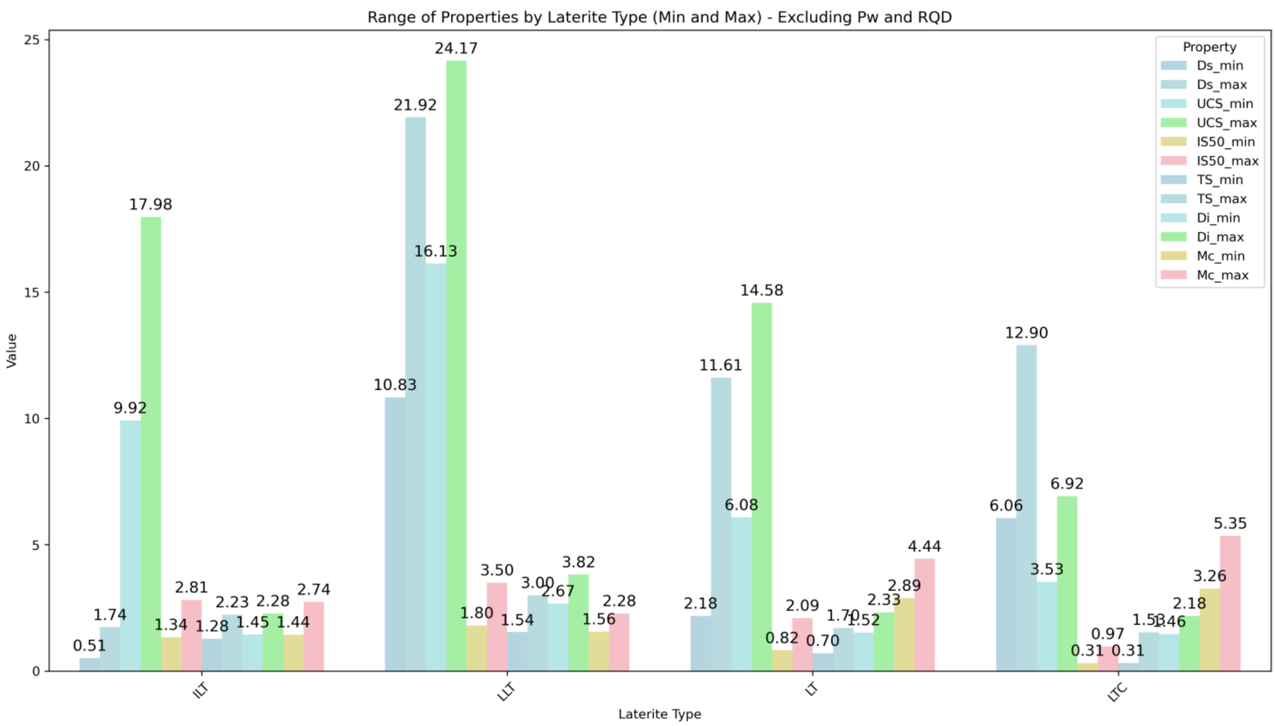
From the graph, the most significant features in laterite classification are:

1. Uniaxial Compressive Strength (UCS) – Highest importance, indicating that rock strength plays a major role in laterite differentiation.
2. Dry Density (Ds) – Strongly related to the compaction and structural integrity of laterites.
3. Density (Di) – Determines the degree of laterization and mineral composition.
4. Point Load Strength Index (IS50) – A direct strength parameter aiding in rock classification.
5. Moisture Content (Mc) – Influences the workability and weathering characteristics of laterite.
6. Tensile Strength (TS) – Moderate impact, affecting fracture resistance of laterite.
7. Porosity (Pw) – Affects fluid retention and permeability, playing a minor role in classification.
8. Rock Quality Designation (RQD) – Lowest importance, suggesting that rock mass integrity has a limited role in laterite classification.

Range Analysis of Laterite Properties

This analysis is to determine the **range (minimum and maximum values)** of various geotechnical and mechanical properties of Laterite rock types. This helps in understanding the **variability and distribution** of different Laterite types based on their key physical parameters.

	Laterite Type	Ds_min	Ds_max	UCS_min	UCS_max	IS50_min	IS50_max	TS_min	TS_max	Pw_min	Pw_max	Di_min	Di_max	Mc_min	Mc_max	RQD_min	RQD_max
0	ILT	0.510000	1.740000	9.920000	17.980000	1.340000	2.810000	1.280000	2.230000	1973.970000	2773.410000	1.450000	2.280000	1.440000	2.740000	23.300000	39.820000
1	LLT	10.830000	21.920000	16.130000	24.170000	1.800000	3.500000	1.540000	3.000000	2338.100000	3479.060000	2.670000	3.820000	1.560000	2.280000	25.340000	49.610000
2	LT	2.180000	11.610000	6.080000	14.580000	0.820000	2.090000	0.700000	1.700000	1547.300000	2450.440000	1.520000	2.330000	2.890000	4.440000	17.310000	36.150000
3	LTC	6.060000	12.900000	3.530000	6.920000	0.310000	0.970000	0.310000	1.530000	1286.830000	2172.480000	1.460000	2.180000	3.260000	5.350000	12.880000	26.070000



- This analysis provides insights into Laterite type variability across different features.
- If some properties have wide or overlapping ranges, it may indicate the need for additional features or more granular classification methods.

User-Input-Based Prediction Process

The prediction function allows users to enter eight geotechnical properties, after which the trained machine learning model classifies the laterite type. The process follows these key steps:

1. Data Collection from the User:
    - The user provides values for the following geotechnical parameters:
        - Dry Density (Ds)
        - Unconfined Compressive Strength (UCS)
        - Point Load Strength Index (IS50)
        - Tensile Strength (TS)
        - Porosity (Pw)
        - Durability Index (Di)
        - Moisture Content (Mc)
        - Rock Quality Designation (RQD)

```
Enter the 8 input properties:
Ds:
```

2. Data Preprocessing:
    - Since the machine learning model was trained on standardized features, the input values are transformed using the same StandardScaler that was used during model training.
    - This ensures consistency and prevents bias due to variations in numerical scales.
3. Prediction Using the Random Forest Model:
    - The processed input is fed into the trained Random Forest Classifier, which predicts the laterite type.
    - The model assigns an encoded numerical label to the input data, which represents a specific laterite category.
4. Decoding and Displaying Results:
    - The encoded label is mapped back to its corresponding laterite category using the Label Encoder.
    - The final prediction is displayed, providing the laterite type based on the input properties.

```
Enter the 8 input properties:
Ds:  0.75
UCS:  15.46
IS50:  2.72
TS:  1.88
Pw:  2542.18
Di:  1.63
Mc:  1.99
RQD:  31.49

◆ Predicted Result:
  - Encoded Label: 0
  - Laterite Type: ILT
```

5. Conclusion

This study successfully implemented a Machine Learning (ML) model for the classification of laterite materials based on their geotechnical properties. The Random Forest classifier achieved an impressive accuracy of 95%, demonstrating its effectiveness in differentiating between Iron-Rich Laterite (ILT), Laterite (LT), Low-Grade Laterite (LLT), and Lateritic Clay (LTC). The analysis revealed that Unconfined Compressive Strength (UCS), Rock Quality Designation (RQD), and Point Load Strength Index (IS50) were the most influential parameters in predicting laterite type.

By leveraging data-driven techniques, this approach significantly reduces the time and effort required for traditional manual classification. The findings highlight the potential of ML-based classification in improving accuracy and efficiency in geotechnical and mining applications.

Applications

The findings of this research can be applied across various fields such as Geology, Construction, Mining, and Environmental Management:

1. **Geology and Construction Industry:**
   - Helps in selecting suitable Laterite materials for roads, foundations, and slope stability.
   - Enables faster and more accurate classification of construction materials.
2. **Mineral Resource and Mining Industry:**
   - Assists in efficient resource classification and mine planning.
   - Enhances excavation strategies by quickly analyzing Laterite properties.
3. **Material Testing and Quality Control:**
   - Streamlines material quality assessment using a data-driven approach.
   - Reduces reliance on laboratory testing by developing an automated classification system.
4. **Environmental and Sustainable Development:**
   - Ensures optimal utilization of Laterite types, reducing environmental impact.
   - Minimizes material waste through efficient classification and selection.